



Assignment Submission Cover Page

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Course Title	HIGHER DIPLOMA IN COMPUTER SCIENCE DIPLOMA IN INFORMATION TECHNOLOGY
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Submission Date:	

1. Convert the binary integer to denary and hexadecimal.

Binary: 1000 1111 1010 1011

- **Hexadecimal Conversion:**

- Group into nibbles: 1000 1111 1010 1011
- Convert each nibble:
 - 1000 = 8
 - 1111 = F
 - 1010 = A
 - 1011 = B
- **Result: 8FAB**

- **Decimal (Denary) Conversion:**

- Calculate weighted sum: $1 \times 2^{15} + 0 \times 2^{14} + 0 \times 2^{13} + 0 \times 2^{12} + 1 \times 2^{11} + 1 \times 2^{10} + 1 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
- = 32768 + 2048 + 1024 + 512 + 256 + 128 + 32 + 8 + 2 + 1
- = **36779**

- **Final Answer:** Denary: **36779**, Hexadecimal: **8FAB**

2. Convert the given ASCII value to binary and hexadecimal

ASCII: Good Luck

- **Step 1: Find Decimal Values:**

- G=71, o=111, o=111, d=100, Space=32, L=76, u=117, c=99, k=107

- **Step 2: Convert each to 8-bit Binary:**

- 71 = 01000111
- 111 = 01101111
- 111 = 01101111
- 100 = 01100100
- 32 = 00100000
- 76 = 01001100
- 117 = 01110101

- 99 = 01100011
- 107 = 01101011
- **Binary Result:** 01000111 01101111 01101111 01100100 00100000 01001100 01110101 01100011 01101011
- **Step 3: Convert each to Hex:**
 - 71 = 0x47
 - 111 = 0x6F
 - 111 = 0x6F
 - 100 = 0x64
 - 32 = 0x20
 - 76 = 0x4C
 - 117 = 0x75
 - 99 = 0x63
 - 107 = 0x6B
- **Hexadecimal Result:** 47 6F 6F 64 20 4C 75 63 6B

3. Complete the table for the MAC address section – 8B EA 9C 59

Hexadecimal	8-Bit Binary
8B	10001011
EA	11101010
9C	10011100
59	01011001

4. Explain why hexadecimal is used to represent colour codes in HTML and CSS.

Hexadecimal provides a concise and human-readable way to represent the binary values that computers use to store colors. A color is defined by three 8-bit values for Red, Green, and Blue (RGB). In decimal, this would be three numbers (e.g., `rgb(255, 0, 128)`). In hexadecimal, each 8-bit value is represented by just two characters (e.g., `#FF0080`). This notation is significantly more compact and easier to read, write, and remember than long binary strings or a series of decimal numbers.

5. Convert the binary number 11101011 to decimal.

- **Binary:** 11101011
 - **Calculation:** $1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
 - $= 128 + 64 + 32 + 0 + 8 + 0 + 2 + 1$
 - **Result:** 235
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6. Convert the octal number 503642 to decimal number.

- **Octal:** 503642
 - **Calculation:** $5 \times 8^5 + 0 \times 8^4 + 3 \times 8^3 + 6 \times 8^2 + 4 \times 8^1 + 2 \times 8^0$
 - $= (5 \times 32768) + 0 + (3 \times 512) + (6 \times 64) + (4 \times 8) + (2 \times 1)$
 - $= 163840 + 0 + 1536 + 384 + 32 + 2$
 - **Result:** 165794
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7. Convert the binary number 11011011101101 to octal and decimal.

- **Binary:** 011 011 011 101 101 (Grouped from right to left)
- **Octal Conversion:**
 - 011 = 3
 - 011 = 3
 - 011 = 3
 - 101 = 5
 - 101 = 5
 - **Result:** 33355
- **Decimal Conversion:**
 - Calculate: $1 \times 2^{13} + 1 \times 2^{12} + 0 \times 2^{11} + 1 \times 2^{10} + 1 \times 2^9 + 0 \times 2^8 + 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
 - $= 8192 + 4096 + 0 + 1024 + 512 + 0 + 128 + 64 + 32 + 0 + 8 + 4 + 0 + 1$
 - **Result:** 14061

8. Convert the decimal number 605 to binary.

- **Decimal:** 605
 - **Repeated Division by 2:**
 - $605 / 2 = 302 \text{ rem } 1$
 - $302 / 2 = 151 \text{ rem } 0$
 - $151 / 2 = 75 \text{ rem } 1$
 - $75 / 2 = 37 \text{ rem } 1$
 - $37 / 2 = 18 \text{ rem } 1$
 - $18 / 2 = 9 \text{ rem } 0$
 - $9 / 2 = 4 \text{ rem } 1$
 - $4 / 2 = 2 \text{ rem } 0$
 - $2 / 2 = 1 \text{ rem } 0$
 - $1 / 2 = 0 \text{ rem } 1$
 - **Read remainders from bottom to top: 1001011101**
 - **Result:** 1001011101
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9. Convert the hexadecimal number 8C5E to octal.

- **Step 1: Hex to Binary.**
 - $8 = 1000$
 - $C (12) = 1100$
 - $5 = 0101$
 - $E (14) = 1110$
 - **Binary:** 1000 1100 0101 1110
- **Step 2: Binary to Octal.** Group into sets of 3 from right: 001 001 100 010 111 110
 - $001 = 1$
 - $001 = 1$
 - $100 = 4$
 - $010 = 2$

- $111 = 7$
 - $110 = 6$
 - **Result: 114276**
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10. Convert the binary number 100011101111001 to hexadecimal.

- **Binary:** 0100 0111 0111 1001 (Grouped into nibbles from right, padded)
 - **Convert each nibble:**
 - $0100 = 4$
 - $0111 = 7$
 - $0111 = 7$
 - $1001 = 9$
 - **Result: 4779**
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11. Convert the decimal number 5372 to hexadecimal and octal.

- **Decimal:** 5372
- **To Hexadecimal ($\div 16$):**
 - $5372 / 16 = 335 \text{ rem } 12 \text{ (C)}$
 - $335 / 16 = 20 \text{ rem } 15 \text{ (F)}$
 - $20 / 16 = 1 \text{ rem } 4$
 - $1 / 16 = 0 \text{ rem } 1$
 - **Result: 14FC**
- **To Octal ($\div 8$):**
 - $5372 / 8 = 671 \text{ rem } 4$
 - $671 / 8 = 83 \text{ rem } 7$
 - $83 / 8 = 10 \text{ rem } 3$
 - $10 / 8 = 1 \text{ rem } 2$
 - $1 / 8 = 0 \text{ rem } 1$
 - **Result: 12374**

12. Convert the hexadecimal number 9A5F to decimal.

- **Hex:** 9A5F
 - **Calculation:** $9 \times 16^3 + 10 \times 16^2 + 5 \times 16^1 + 15 \times 16^0$
 - $= (9 \times 4096) + (10 \times 256) + (5 \times 16) + (15 \times 1)$
 - $= 36864 + 2560 + 80 + 15$
 - **Result:** 39519
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13. (a) Translate English to Propositional Logic

- (i) $R \leftrightarrow S$
- (ii) $(S \rightarrow R) \wedge (R \rightarrow S)$
- (iii) $R \leftrightarrow S$
- (iv) $H \wedge HA$
- (v) $C \vee D$

(b) Translate Propositional Logic to English

- (i) If John is eating, then he is not hungry.
 - (ii) John is eating and he is not hungry.
 - (iii) It is not true that if John is hungry then he is not eating. (John is hungry and he is eating).
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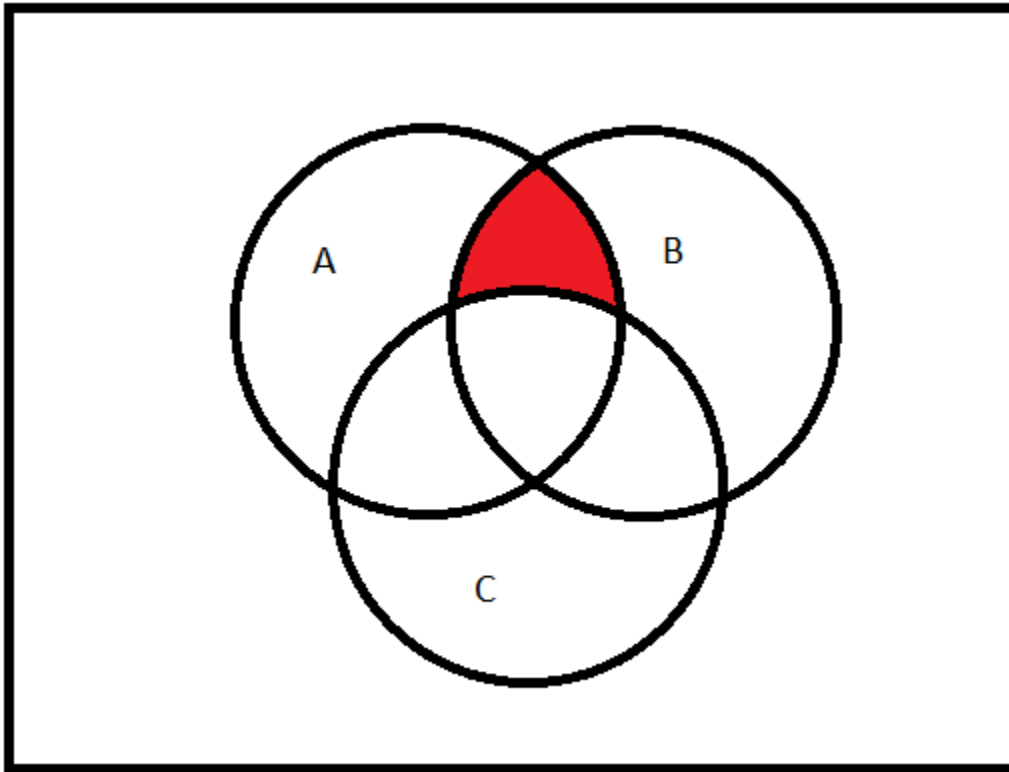
14. Read the Venn Diagram

Based on standard Venn diagram interpretation:

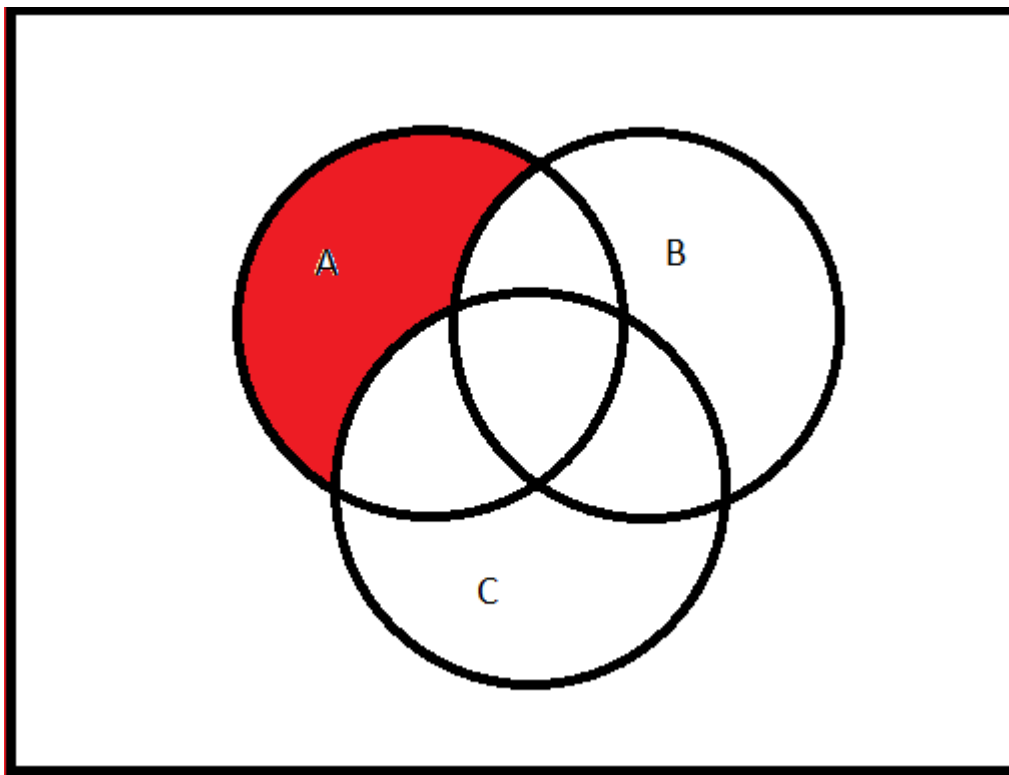
- (i) $B' : \mathbf{a, f, e, j}$
- (ii) $A \cup B : \mathbf{a, i, c, d, g, f}$
- (iii) $A \cap B : \mathbf{i, d}$
- (iv) $(A \cup B)' : \mathbf{j, h, e}$
- (v) $(A \cap B)' : \mathbf{a, c, f, g, j, h, e}$
- (vi) $A - B : \mathbf{a, f}$
- (vii) $B - A : \mathbf{c, g}$

15. Draw Venn Diagrams

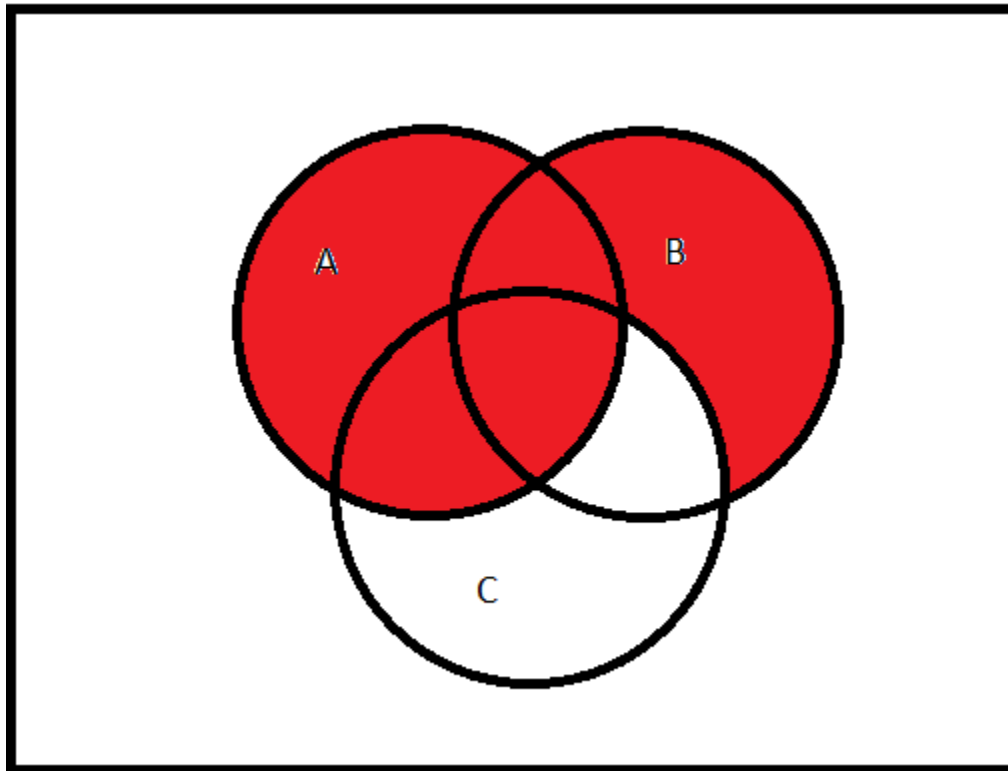
(i) $A \cap B \cap C'$: The overlapping region of circles A and B, excluding any part of circle C.



(ii) $A \cap B' \cap C'$: The part of circle A that does not overlap with circle B or circle C.



(iii) **$A \cup (B \cap C)$** : All of circle A combined with the part of circle B that is outside circle C.



16. Construct Truth Tables

(a) Prove that $p \vee q \equiv \neg(\neg p \wedge \neg q)$

P	Q	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$	$\neg(\neg P \wedge \neg Q)$	$P \vee Q$
T	T	F	F	F	T	T
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	T	F	F
<i>The equivalence is proven as the last two columns are identical.</i>						

(b) $\sim[(p \wedge q) \rightarrow (r \wedge q)]$

p	q	r	$p \wedge q$	$r \wedge q$	$(p \wedge q) \rightarrow (r \wedge q)$	$\sim[\dots]$
T	T	T	T	T	T	F
T	T	F	T	F	F	T
T	F	T	F	F	T	F
T	F	F	F	F	T	F
F	T	T	F	T	T	F
F	T	F	F	F	T	F
F	F	T	F	F	T	F
F	F	F	F	F	T	F

(c) Show that $[(p \vee q) \wedge \neg p] \rightarrow q$ is a tautology.

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$	$[(p \vee q) \wedge \neg p] \rightarrow q$
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T
<i>The final column is always True, so it is a tautology.</i>					

17. Multiplying Polynomials

$$\begin{aligned}
 & \text{(i) } (2a - 8b)(6a - 8b) \\
 &= 12a^2 - 16ab - 48ab + 64b^2 \\
 &= \mathbf{12a^2 - 64ab + 64b^2}
 \end{aligned}$$

$$\begin{aligned}
 & \text{(ii) } (7m - 5n)(2m + 5n) \\
 &= 14m^2 + 35mn - 10mn - 25n^2 \\
 &= \mathbf{14m^2 + 25mn - 25n^2}
 \end{aligned}$$

18. Probability

(a) P(Sum of 8 with two dice)

- Favorable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2) → 5 outcomes.
- Total outcomes: 36.
- $P(\text{Sum}=8) = 5/36$

(b) Toss coin then die

- **Sample Space S:** {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}
- $P(T, 3)$: 1 favorable outcome (T3) / 12 total outcomes.
- $P(T, 3) = 1/12$

(c) P(Diamond or King from 52-card deck)

- $P(\text{Diamond}) = 13/52$
- $P(\text{King}) = 4/52$
- $P(\text{King of Diamonds}) = 1/52$
- $P(D \text{ or } K) = 13/52 + 4/52 - 1/52 = 16/52 = 4/13$

(d) Bag: 6 Red, 8 Yellow, 4 Green. Total = 18.

(i) $P(\text{Yellow}) = 8/18 = 4/9$

(ii) $P(\text{Green}) = 4/18 = 2/9$

(iii) $P(\text{Yellow or Green}) = 8/18 + 4/18 = 12/18 = 2/3$

19. Find Mean and Standard Deviation

Mass(kg)	Mid-point (x)	Freq (f)	f*x	(x - μ)	(x - μ) ²	f*(x - μ) ²
60-62	61	5	305	-6.82	46.5124	232.562
63-65	64	18	1152	-3.82	14.5924	262.6632
66-68	67	42	2814	-0.82	0.6724	28.2408
69-71	70	27	1890	2.18	4.7524	128.3148
72-74	73	8	584	5.18	26.8324	214.6592
Total		100	6745			866.44

- **Mean (μ)** = $\Sigma(f*x) / \Sigma f = 6745 / 100 = 67.45$ kg
- **Variance (σ^2)** = $\Sigma[f*(x - \mu)^2] / \Sigma f = 866.44 / 100 = 8.6644$
- **Standard Deviation (σ)** = $\sqrt{8.6644} \approx 2.94$ kg